Does Grounding *Ground* or *Require* Necessity?

Abstract

Where does necessity come from? This paper contrasts two accounts of the relation between grounding and metaphysical necessity. On the *groundist* account, developed most systematically by De Rizzo (2020), grounding facts are the source of necessity facts. On the *constitutivist* account defended here, grounding is constitutively necessitating: each grounding fact requires, in virtue of its metaphysical constitution, the corresponding necessitation fact. I argue that these views are incompatible once fundamentality is understood generatively, as the *ordo generationis*. This conception motivates a refined grounding-to-fundamentality principle on which first grounds determine generative priority, and a constitution-to-anti-fundamentality principle on which constitutive requirements cannot point to what is metaphysically posterior. Together, these principles undermine the claim that grounding is the generative source of necessity. I further argue that this result survives fundamentality pluralism by defending a single global availability order constraining all genuinely generative relations.

[*Keywords*: Fundamentality, Generation, Constitution, Dependence, Necessitation]

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1. Introduction

I want to contrast two pictures of the relation between grounding and metaphysical necessity, show that they are incompatible, and argue that we should prefer one over the other. The argument is made against the background of a *generative* conception of fundamentality, as well as a *predicative* theory of metaphysical generation.

On one picture, grounding is the *ultimate source of necessity*: necessity facts are ultimately explained by grounding facts. Call this the *groundist view of necessity*. De Rizzo (2020) offers the most systematically developed groundist view, and is the one I will take here as the representative.[1]

[1] There are some antecedent suggestions to take grounding as involved in the explanation of necessity. Rosen (2006) thought of the 'standard' conception of metaphysical necessity as taking a proposition to be necessary when it holds in every broadly possible world that shares the actual *laws of metaphysics*. These include principles specifying how derivative entities are generated from or grounded in the basic array. Later (2010), he tentatively suggested that necessities may be taken to be the consequences of the essential truths *together* with some basic grounding laws. Rosen's suggestions take grounding laws as a possible additional source of necessity, though they do not yet amount to the claim that every necessity is fully grounded in a grounding fact. Relatedly, Kment (2014, pp. 187–188) defines a proposition to be metaphysically necessary iff it is true at every world that matches actuality

On the other picture, *grounding is constitutively necessitating*: apart from being explanatory, grounding is a *necessitating connection.* So, if ψ grounds φ, the very constitution of this fact requires that ψ necessitates φ. Call this the *constitutivist* view of grounding.

Both pictures are motivated abductively. I argue that the groundist and constitutivist explanations are incompatible. The argument is initially stated using the familiar connection between grounding and relative fundamentality; but I later refine that connection through a generative conception of fundamentality, and a predicative conception of generation. This conception is also used to motivate the uniqueness of the fundamentality order presupposed by the argument. In the resulting picture, constitutivism is paired with the generative-predicative conception of fundamentality, and this is argued to recapture widely held intuitions about the *plexological family* (as Bennett, forthcoming, calls it)—including grounding, fundamentality, generation and building.

Before explaining the ideas further, let us introduce some notation and disclaimers:

1. 'Δ' and other uppercase Greek letters denote pluralities of facts; pluralities may be unary. '$\bigwedge \Delta$' is the conjunction of the facts in Δ.

2. I will write as if grounding were a *relation* between *facts*, or pluralities thereof. (With some patience and ingenuity, my claims can be translated to a view according to which grounding is an *operation*, and to a view which remains neutral as to the ontology and only posits a sentential operator.) '$\Delta > \varphi$' says that Δ fully grounds φ, full grounding is at least asymmetric and transitive; '$>_i$' is *immediate* full grounding, which is asymmetric and *in*transitive.

3. All grounding claims are to be read factively—if $p > q$, then both p and q obtain.[2]

4. There is some disagreement on whether grounding *is* metaphysical explanation or merely *backs* metaphysical explanations (Raven, 2015); I remain neutral on this issue.

5. I use 'necessity' and '$\Box$' to mean *metaphysical necessity.*

6. *Requirements of metaphysical constitution:* '$\varphi \underset{C}{\Rightarrow} \psi$' means that φ requires, in virtue of its metaphysical constitution, ψ.

7. *Metaphysical fundamentality:* '$\psi \gg \varphi$' means that ψ is more metaphysically fundamental than φ.

with respect to its metaphysical laws; these laws systematically govern grounding but are not generally grounding propositions themselves.

[2]De Rizzo's account assumes a non-factive extension of grounding (see §3, below). While that is required for his more ambitious explanatory programme, it is irrelevant to my counterargument, which requires only true instances of **EN**.

The two views to be examined here assume that grounding necessitates, i.e., that the following holds for all p and Δ:

Necessitation If $\Delta > p$, then $\Box(\bigwedge \Delta \rightarrow p)$.

This thesis is not uncontroversial.[3] However, those who reject **Necessitation** already possess an independent objection to De Rizzo's account; the present argument grants its starting point.

The two views not only accept, but moreover strengthen, **Necessitation**. Groundism strengthens it to the claim that the necessitation fact is grounded in the grounding fact; constitutivism strengthens it to the claim that the grounding fact requires, in virtue of its constitution, the necessitation fact. Then, the two opposing views at the centre of this paper are as follows.

Necessity Constitutivism (NC): The necessitating character of grounding is constitutive of it: each grounding fact entails, because of its very metaphysical constitution, a corresponding necessitation fact:

$$\text{If } \Delta > p, \text{ then: } (\Delta > p) \underset{C}{\Rightarrow} \Box\left(\bigwedge \Delta \rightarrow p\right)$$

Explanatory Necessitation (EN): The fact that the grounds necessarily imply their groundees is itself grounded in their grounding connection:

$$\text{If } \Delta > p, \text{ then: } (\Delta > p) > \Box\left(\bigwedge \Delta \rightarrow p\right)$$

In this paper, I will compare the merits of these opposed views, and argue for constitutivism. First, I will argue that they *are* opposed: given some reasonable background assumptions, the truth of both leads to contradiction (§2). Then, I will explain DeRizzo's case for **EN** (§3). After that, I will argue in favour of the necessity-constitutivist approach to grounding (§4). Finally, I will motivate the background principles from the generative-predicative conception and respond to some potential objections (§§5–6).

2. The Argument

For any *grounding fact* G, i.e., a fact of the form (for some p, Δ):

$$G = (\Delta > p),$$

[3] *Grounding contingentists* (e.g., Leuenberger, 2014; Skiles, 2015) argue that A can ground B in world w_1, but fail to ground B in world w_2 because of a change in background conditions. So, they reject **Necessitation** as presented, but accept a version where background conditions are included in the antecedent. *Grounding indeterminists* (e.g., Wygoda Cohen, 2024; Pearce, 2025) go further: they claim that a total ground might have failed to ground what it actually grounds—even the addition of background conditions cannot rescue **Necessitation**.

there is, according to **Necessitation**, an *associated necessitation fact* N:

$$N = \Box(\bigwedge \Delta \rightarrow p).$$

In this notation, groundism's **EN** universally quantifies the following claim:

$$G > N \tag{1}$$

Now, if grounding explanations track metaphysical fundamentality,

$$G \gg N. \tag{2}$$

But constitutivism says that the grounding fact G, of its very constitution, requires the modal connection N:

$$G \underset{C}{\Rightarrow} N. \tag{3}$$

Then, if the constitution of a fact cannot require a less fundamental fact,

$$\neg(G \gg N). \tag{4}$$

Contradiction.

As we see, I need two premisses:

Grounding to Fundamentality (GTF): If φ is fully grounded in Δ, then Δ is more fundamental than φ:

$$\text{If } (\Delta > \varphi), \text{ then } (\Delta \gg \varphi)$$

Constitution to Anti-Fundamentality (CTAF): If, because of its very constitution, φ requires ψ, then φ is *not* more fundamental than ψ:

$$\text{If } (\varphi \underset{C}{\Rightarrow} \psi), \text{ then } \neg(\varphi \gg \psi).$$

Then, **Grounding to Fundamentality** validates the transition from (1) to (2), and **Constitution to Anti-Fundamentality** validates the transition from (3) to (4). Formally:

1.	$G = (\Delta > p)$.	
2.	$N = \Box(\bigwedge \Delta \rightarrow p)$.	
3.	$G \underset{C}{\Rightarrow} N$.	**NC**
4.	$\therefore \neg(G \gg N)$.	**CTAF**, 3
5.	$G > N$.	**EN**, 1, 2
6.	$\therefore G \gg N$.	**GTF**, 5
7.	$\bot$.	

The accepted datum for both the constitutivists and the groundists is the necessitating aspect of grounding: $\Box(G \rightarrow N)$, for all grounding facts and associated necessitation facts. Groundism explains this systematic connection through **EN**: thus, G grounds N, and from **Necessitation**, $\Box(G \rightarrow N)$. Meanwhile, constitutivism explains the systematic connection by claiming that (i) $G \underset{C}{\Rightarrow} N$, and (ii) constitutive requirement entails necessitation: if G requires N because of its constitution, $\Box(G \rightarrow N)$. The question now is which explanation is better. So, my argument in favour of the constitutivist picture will be abductive.

Also, note that the argument only requires the priority constraint that *Necessitation is not metaphysically posterior to grounding*—locally:

$$\neg(G \gg N). \tag{PC}$$

I came to (PC) by believing in the constitutive account (which entails (PC) through **CTAF**); there may be alternative routes.

Importantly, what I presented in this section is the simplest form of the argument. As we shall see in §5.2, **GTF** faces counterexamples arising from multiply grounded facts. I will therefore replace it with a more discriminating generative principle. The new principle is motivated by a conception of generation and fundamentality; this conception also motivates the assumption, implicit in the argument, that there is a unique order of metaphysical fundamentality. The resulting version of the argument preserves the present tension while making explicit the sense in which grounding must be prior to necessity if it is genuinely to be its metaphysical source.

3. De Rizzo's Groundist Approach to Necessity

3.1. From Necessitation to Explanatory Necessitation

The starting assumption of De Rizzo (2020) is **Necessitation**. Then, his central move is to transform this merely modal consequence of grounding into an explanation of the modal fact itself. The fundamental principle is then *Explanatory Necessitation*:

EN If p is grounded in Δ, then this fact grounds the fact that Δ necessitates p:

$$\text{If } \Delta > p, \text{ then: } (\Delta > p) \text{ grounds } \Box(\bigwedge \Delta \rightarrow p).$$

So, grounding does not merely imply modal connections: it *explains* those modal connections. This is the heart of the proposal.[4]

[4]De Rizzo wants **EN** to explain necessities such as $\Box(p \rightarrow p \vee q)$, even when p is false; he therefore interprets 'because' *non-factively* for the purposes of **EN**. As announced before, here I am restricting the discussion to facts.

EN naturally generates necessities of the form: $\Box(\Delta \rightarrow p)$. But many necessities do not have that form. What about propositions of the form $\Box p$—like $\Box$(Socrates = Socrates)? Moreover, the asymmetry of grounding creates problems for some tautological conditionals. For instance, we should not say: $p > p$, because grounding is irreflexive and asymmetric. So, **EN** cannot straightforwardly explain $\Box(p \rightarrow p)$ by appealing to p grounding itself.

3.2. Zero grounding as explanation of other necessities

This leads De Rizzo to a substantive theoretical commitment: *zero grounding*. Following Fine, De Rizzo distinguishes between ungrounded propositions and zero-grounded propositions. Let $\varnothing$ represent the empty plurality of truths. Then, '$\varnothing > p$' means that p *is grounded*, but grounded *in no propositions*—or even that p 'depend[s] on nothing at all' (De Rizzo, 2020, p. 643) or owes its truth 'to no circumstance at all' (p. 644).

To understand this, the usual resource is the analogy of *Fine's machine*. Imagine a machine that accepts propositions as inputs and produces their groundees as outputs. Feed it p, q, and perhaps it outputs $p \wedge q$. Feed it nothing, and if it nevertheless outputs r, then, r is zero-grounded. By contrast, if some proposition s never emerges from the machine at all, it is ungrounded. So, to be *zero-grounded* is different from being *ungrounded*.

An ungrounded proposition is simply one for which there is no reason why it obtains: it is a brute or fundamental fact. Meanwhile, *zero-grounded truths* depend upon the weakest possible condition: the empty plurality of truths. Hence, a zero-grounded truth is one which obtains 'no matter what' (p. 643). And that immediately connects zero-grounding to necessity, for if $\varnothing > p$, then **Necessitation** gives something equivalent to $\Box p$. Thus, zero-grounded truths are necessarily true. This distinction is fundamental to De Rizzo's account, as it permits him to reject the inference from there being no ordinary grounds for p to p being explanatorily brute. But if propositions are individuated finely, so that p and $\varnothing \rightarrow p$ are genuinely different propositions, **EN** alone is insufficient. Hence, De Rizzo introduces a separate principle, Explanatory Necessitarianism+:

EN$^+$ If p is grounded in $\varnothing$, then this fact grounds the fact that $\Box p$.

Thus, a zero-grounded proposition is necessary *because* it is zero-grounded.

De Rizzo regards the following as plausible zero-grounded truths: *logical truths*, *identity and difference truths*—such as Socrates≠Eiffel Tower—, *modal-logic validities*, *essential and natural-kind necessities*, and Kripkean *a posteriori* necessities.

At this stage we get the basic theory: p's being zero-grounded explains $\Box p$, and ordinary grounding relations, $\Delta > p$, explain corresponding strict implications: $\Box(\Delta \rightarrow p)$. But De Rizzo thinks that this

still fails to capture important relations between necessities themselves, and introduces two additional principles.

3.3. Further principles of the theory

First, strict implication is monotonic: if $\Box(p \rightarrow q)$, then $\Box((p \wedge r) \rightarrow q)$. Adding assumptions to the antecedent cannot destroy a necessary implication. De Rizzo thinks that there is not merely an entailment here, but also an explanation: $\Box((p \wedge r) \rightarrow q)$ holds *because* $\Box(p \rightarrow q)$. He therefore proposes:

ME: If $\Box(\bigwedge \Delta \rightarrow p)$, then for every $\Gamma \supseteq \Delta$: $\Box(\bigwedge \Delta \rightarrow p)$ grounds $\Box(\bigwedge \Gamma \rightarrow p)$.

The richer antecedent contributes nothing to the necessity of the implication. The real explanatory work is already being done by the thinner conditional.

The fourth principle is inspired by the logic of grounding, where a natural intuition is that the truth of a logically complex proposition is grounded in appropriate truths about its constituents. De Rizzo proposes extending this idea, so that a complex proposition may owe not merely its truth, but its necessity, to the necessity of its components. Consider: $\Box(p \vee q)$ where $\Box p$. If p is necessary, then $p \vee q$ is necessary because p is necessary. The general principle is:

ECI: If $\Delta > p$ and $\Box \bigwedge \Delta$, then: $\Box p$ because the necessities of the propositions in Δ.

So if p because q, r, s and $\Box q, \Box r, \Box s$, then: $\Box p$ because $\Box q, \Box r, \Box s$.

Putting **EN**, **EN**$^+$, **ME** and **ECI** together, the paper ultimately proposes the following explanatory architecture:

$$
\begin{array}{lcccl}
\text{Level 1:} & \varnothing > p & \xrightarrow{EN^+} & \Box p & \\
\text{Level 2.} & \Delta > p & \xrightarrow{EN} & \Box(\bigwedge \Delta \rightarrow p) & \\
\text{Level 3:} & \Box(\Delta \rightarrow p) & \xrightarrow{ME} & \Box(\Gamma \rightarrow p), \text{ with } \Delta \subseteq \Gamma & \\
\text{Level 4:} & \Box q_1, \ldots, \Box q_n & \xrightarrow{ECI} & \Box p & \text{when } \{q_1, \ldots, q_n\} > p.
\end{array}
$$

The first two arrows *inject* necessity facts into the grounding explanatory network; **ME** and **ECI** then *propagate* necessity through that network. In what follows, I identify this account with *groundism about necessity.*

3.4. Evaluation

I will be putting **EN**$^+$ to the side and focus on **EN**. To explain why, remember that the proposed explanatory schema justifying **EN**$^+$ is as follows:

(i) Many, if not all, of the necessities that seem ungrounded are actually grounded in $\varnothing$;

(ii) $\varnothing$ is the weakest possible condition on the world, and therefore obtains regardless what else transpires;

(iii) if p is grounded in something that obtains 'no matter what' (De Rizzo, 2020, p. 643), **Necessitation** entails that p also obtains 'no matter what'.

(iv) Therefore, if p is grounded in $\varnothing$, this explains p's necessity.

But this schema breaks down at various steps.

First (as Haderlie & Miller, forthcoming, §3.1, argue), $\varnothing$ is already characterised by necessity (see (ii) in the schema), which would make **EN**$^+$ unable to truly source necessity.

Second, *pluralities* do not obtain: *true propositions* or *facts* obtain. Thus, the defender of zero grounding cannot say that the empty plurality of truths obtains (necessarily or not). The claim must instead be that *the elements of the empty plurality* obtain, or that *their conjunction* obtains. But this formulation exposes the issue: there is no metaphysical condition constituted by *no propositions*, by 'no circumstance at all'—even if $\bigwedge \varnothing$ functions *algebraically* as $\top$ (*cf.* Lo *et al.*, forthcoming)—against (ii) in the schema.

Third, and graver, on a Boolosian conception of plural reference (Boolos, 1984), there are no empty pluralities to begin with: pluralities are not *further* objects, *over and above* their members: where there are no members there is no plurality. 'The Beatles' does not refer to a *further* object: it is a notational hypostatisation of how we plurally refer to John, Paul, George, and Ringo. Sets are different from their members; that is why there can be a set with no members. No analogous explanation is available for an empty plurality. Then, even if the empty plurality is legitimate as an algebraic construction, that is not sufficient to regard it as an admissible metaphysical ground—against (i).

Without **EN**$^+$, the only principle remaining in the theory that explains necessities from non-necessities is **EN**, and that is why it is the main focus of my discussion: constitutivism is consistent with the claim that some necessities are grounded in other necessities.

Now, **EN** only says that G-facts explain N-facts: given that $\Delta > p$, the necessitation: $\Box(\bigwedge \Delta \rightarrow p)$ is explained by p being grounded in $\bigwedge \Delta$. But how is *that* explanation supposed to go? *Where is the explanation?* In general, the mere claim that there is an explanatory connection between p and q does not give us the actual explanation (*cf.* Daly, 2023).

So, what is the motivation of **EN**? I assume it is that, together with De Rizzo's other principles, it provides a coherent theory of necessity. The theory is then presented in the hopes that 'the outline of the view suffices to make it a serious option' (De Rizzo, 2020, p. 640). The idea seems to be that, by examining the principles of the theory, we can evaluate its explanatory power and other theoretical virtues. For example, a fundamentalistic Ockham's razor: If all necessity facts are ultimately grounded in grounding

facts, necessity is not fundamental, and we have reduced our list of fundamental commitments. Thus, ultimately, **EN** would be motivated abductively: accepting it, along with a theoretical package, provides a plausible theory of the grounding of necessity, and reduces our fundamental commitments.

4. Why Grounding is Constitutively Necessitating

EN adds a grounding-explanatory arrow from each G-fact to its N-correlate. Constitutivism rejects this. The thought is that the grounding fact does not *first* obtain as a modally neutral fact and *then* generate its necessary connection to the groundee—rather, its being a grounding fact requires it to *already* incorporate the relevant modal determination.

The constitution in question is metaphysical, and so I assume that it entails necessitation: if $\varphi \underset{C}{\Rightarrow} \psi$, then $\Box(\varphi \rightarrow \psi)$—necessarily, if φ obtains, then everything that its constitution requires also obtains. Apart from that, there are different ways in which to understand this constitution. So, if $\varphi \underset{C}{\Rightarrow} \psi$, this may be because φ has a *metaphysical analysis* in terms of ψ, or because φ has its *real definition* or *essence* in terms of ψ, or because φ *just is* ψ (possibly in conjunction with something else). But note: the notion need not be that φ contains ψ as a mereological part. Thus, I claim that $[\forall x(Fx)] \underset{C}{\Rightarrow} [Fa]$, but I am not committed to $[Fa]$ being a mereological part of $[\forall x(Fx)]$.

Now, while De Rizzo provides considerations making **EN** a natural extension of **Necessitarianism**, he has not presented an independent metaphysical account of why the explanatory direction should run from grounding connections to their associated necessitation facts. I aim to provide an independent account explaining the direction that, I think, the explanatory arrow actually takes.

4.1. Determination and nothing-over-aboveness require necessitation

Metaphysical grounding is supposed to be a particularly stringent form of determination. If ψ fully grounds φ, there is supposed to be no further metaphysical latitude concerning φ once ψ is fixed. This is what I assume to be behind all the familiar phrasings, such as that grounded entities are 'nothing over and above' their grounds, that the former obtain 'in virtue of' the latter, or that the grounds 'metaphysically determine' their groundees.

Thus, grounding would seem to be an extremely *tight*, extremely intimate, kind of connection—one almost 'just shy of identity'. This, as Rydéhn (2022, p. 733) says, is a recurring theme in the grounding literature: 'Due to this, a correct grounding explanation is supposed to provide a form of understanding and illumination which is simply not attainable in other kinds of explanations.' Also, Fine (2001, p. 15): '[T]he relation of ground is distinguished from [other explanatory connections] by being the tightest such connection'. And Rosen (2017, p. 281): 'If the explanans can occur with or without the explanandum,

then the explanans is clearly "something over and above"'.

And then, if the full grounds could obtain while their groundee failed to obtain, then those grounds would leave something about their groundee metaphysically *unsettled*. And it is difficult to see how a relation so *tight* as to be just shy of identity could possibly be non-necessitating: the non-necessitating grounding of φ by ψ would mean that φ *is* something over and above ψ, as ψ can obtain *without* φ obtaining.

So, grounding is a very tight relation in this sense, and part of such tightness is the necessity of the connection: part of what makes grounding 'just shy of identity' is that it does not connect grounds to groundees contingently. If this is correct, any adequate metaphysics of grounding must explain why full grounding carries this modal force.[5]

This motivates the necessitarian view of grounding: contingent grounding would not provide the required tightness. On the groundist picture, the necessary connection obtains because grounding grounds it. As we now turn to see, on the constitutivist picture, the modal force is best understood as constitutive of metaphysical grounding, rather than as a consequence externally generated from already constituted grounding facts. So, the sort of metaphysical determination that grounding is supposed to *be* is, in part, constituted by its modal rigidity.

4.2. Necessitation separates grounding from other explanatory relations

There are numerous explanatory and determination relations. *Causal* determination is the relation that obtains between the existence or features of an object, or the occurrence of an event, and its causes. *Nomic* determination is the determination of an event by the laws (along with the initial and/or boundary conditions). *Logical* determination is the determination of a proposition or formula by the premisses of a valid argument which concludes with it. And grounding is posited to be a certain *metaphysical* kind of determination: the determination of a fact by its grounds. All of these relations are explanatory: one can explain why some object is the way that it is in terms of its causes, why some event occurs in terms of how the laws lead to it, why some proposition follows (or is true) in terms of what premisses entail it (and their truth). And of course, grounding theorists believe that grounds explain what they ground.

But note: mere explanatoriness does not distinguish among these kinds of determinations—all of

[5] Rydéhn (2022) motivates cases of grounding that are not as intimate—thus, ψ *opaquely grounds* φ just in case ψ grounds φ without occurring in the essence or metaphysical analysis of φ. However, Rydéhn seems to assume that *even* opaque grounding is necessitating: 'A further possible worry is that once we sever any necessary connection between grounding and relations like metaphysical analysis or essential involvement, it is hard to see what distinguishes grounding from mere metaphysical necessitation' (p. 743). My argument here is only that 'tight' grounding is necessitating; even if not all grounding is tight, still most of it is, and it does not follow that tight grounding's being necessitating is not worthy of explanation.

them are explanatory. So, *being explanatory* underdetermines metaphysical grounding. And I claim that the specific *modal force* is an important part of what individuates grounding from neighbouring explanatory relations.

However, groundists about necessity cannot appeal to the same idea, given the argument presented above (§2) and refined below (§5.5). That is: the groundist, who wants G to be the generative source of the necessity N, cannot simultaneously use N's constitutive availability to characterise G, given the assumptions defended in the rest of the paper.

So, *what grounding is* must be capable of underwriting the necessitating aspect of all grounding connections. But mere explanatoryness cannot do that—because, as we have seen, there are other explanatory and determination relations, and not all of them are metaphysically necessitating.

For this individuation, I would not rely on the pure logic of grounding, because it may turn out to have non-standard models. Indeed, there is a real danger that purely formal axioms underdetermine its nature: even among those who think of ground as unified, there are debates as to its elementary logical form—its well-foundedness, transitivity, reflexivity, or even its adicity (Bliss & Trogdon, 2024). And more 'material' postulates may not completely distinguish it from, e.g. causation (Wilson, 2018).

This, I think, is why Rosen (2010, p. 118) claims that obeying **Necessitation** 'is one respect in which [grounding] differs from causal and other merely nomic forms of determination'. Later, Rosen (2017, p. 281) also argued that 'in the probabilistic case it's evident that the explanandum is not metaphysically grounded in the explanans', *precisely because* 'Γ metaphysically grounds A only if Γ entails A'.

Fine (2012, p. 38) thinks that each basic species of necessity—metaphysical, natural and normative, on his view (Fine, 2002)—is associated with a different species of grounding:

> In each of the above cases, there would appear to be some sort of modal connection between explanandum and explanans. [...] it is arguable that the "force" or "strength" of the modal operator is different in each case.
>
> [...] In addition to the modal connection, there would also appear to be an *explanatory* or *determinative* connection. [...]
>
> We may call an in-virtue claim a statement of *ontological* or *metaphysical ground* when the conditional holds of metaphysical necessity [...]

After this, he considers the following question (pp. 39–40):

> whether each of these explanatory relations should be defined in terms of a single generic relation. Thus it might be thought that "metaphysical ground" should be defined by:

> the fact that A grounds the fact that B iff the fact that B obtains in virtue of the fact that A (in the generic sense) and it is a metaphysical necessity that if A then B,
>
> and similarly for the other cases, but with another modality in place of metaphysical necessity. It might, on the contrary, be thought that each basic modality should be associated with its "own" explanatory relation [...]
>
> I myself am inclined to favour the latter view.

So, on this Finean view, we have three elements: (i) the basic modalities: $\Box_{\text{Met}}$, $\Box_{\text{Nat}}$, and $\Box_{\text{Nor}}$; (ii) the explanatory relations: $\mathit{Expl}_{\text{Met}}$, $\mathit{Expl}_{\text{Nat}}$, $\mathit{Expl}_{\text{Nor}}$ (or a generic relation: Expl); and (iii) each grounding relation, defined by the corresponding explanatory relation and modality. For metaphysical grounding, the proposed definition would be:

$$\varphi <_{\text{Met}} \psi \quad \text{iff}_{\text{df.}} \quad \text{(i) } \text{Expl}_{\text{Met}}(\varphi,\psi), \text{ and (ii) } \Box_{\text{Met}}(\varphi \rightarrow \psi).$$

Correia (2024, pp. 2-3) treats modal force as an internal criterion for characterising the canonical metaphysical grounding relation; so, satisfying **Necessitation** is a way of cashing out the metaphysicality of canonical grounding (with fundamentality as another route). Thus, he proposes that 'for a notion of grounding to be metaphysical is for it to satisfy [**Necessitation**] or [**GTF**] or both'. A bit later on (p. 4), Correia writes that 'A further feature of the canonical notion of grounding is that it is explanatory'. His account is therefore closely related to Fine's proposal: grounding is either explanatory and necessitating, or explanatory and fundamentality-inducing, or both. In other work (Correia, 2021b), he also suggests defining strict full grounding in terms of necessity and relative fundamentality.

Other authors also think that, without being constitutively necessitating, it is unclear what distinguishes grounding from other explanatory relations, particularly causal, nomological or logical ones. And that is why many philosophers introduce grounding as being partly, but crucially, characterised by its link to necessity, given by **Necessitation**. We have seen Rosen, Fine and Correia; other examples are Bennett's (2017, p. 50) argument for **Necessitation** as part of what defines *building relations*, of which grounding is a species; or Trogdon's (2013) explanation of the truth of **Necessitation** as following from the fact that grounding facts depend on modal truths.

In sum, different explanatory relations are determinating and explanatory, and some have resembling logical forms. If part of what distinguishes grounding among other explanatory relations is that grounding is constitutively metaphysically necessitating, constitutivism provides a plausible individuating scheme: the very constitution of grounding facts requires the grounds to necessitate their groundees with metaphysical—not causal or purely logical—force. So, constitutivism provides an elegant internal

explanation for why the grounding fact holds: part of what makes it a *grounding* fact is that it carries this absolute modal guarantee.

4.3. Constitutivism and the explanatory power of groundism

Groundism claims in **EN** that G-facts grounding-explain N-facts. However, not all metaphysical explanations are grounding explanations. Constitutivism offers a rivalling explanation, where G-facts explain N-facts in another, precise sense.

For example, in Glazier's notion of *essentialist explanation* (Glazier, 2017), a fact is explained by the very nature of an object to which the fact is essential. Why does {Socrates} contain Socrates as a member? Because containing Socrates lies in the very nature of that specific set. Now, ψ obtaining because of the essence of φ is one way in which it can be that $\varphi \underset{C}{\Rightarrow} \psi$, but there are other theoretical possibilities (such as metaphysical analysis or 'just-is' statements, assuming all of these ultimately differ). So as to remain neutral, let me introduce the term 'constitutive explanation'.

A *constitutive explanation* of a fact φ says that φ obtains *because* it is required by the constitution of another fact that also obtains. Then, the 'because' in a constitutive explanation denotes a *non-generative* kind of explanation (Wallner, 2022)—in particular, *not* grounding explanation. Constitutive explanations of facts are special cases of constitutive explanations in general. Thus, I may explain why is there a shrimp on my plate by noting that there is a shrimp taco on my plate, and that the constitution of this taco requires the shrimp (among other things). Constitutive explanations explain the occurrence of something by the occurrence of another thing whose constitution requires the *explanandum*.

On the standard picture (to be refined below, §5), the full-grounding-*explanans* is always more fundamental than its full-grounding-*explanandum*. Constitutive explanation is not related to fundamentality in the same way (Wallner, 2022, p. 450): the constitutive-*explanans* need not be more fundamental than its constitutive-*explanandum*. Thus, if there is a shrimp in my plate because$_{\text{constitutive}}$ there is a shrimp taco in my plate, it does not follow that the shrimp taco is more fundamental than the shrimp.[6]

If we accept constitutive explanation, we can formulate the following:

Constitutive-explanatory Necessitation (CEN): The grounds necessitate their groundees because (i) the groundees are grounded in the grounds, and (ii) grounding connections are constitutively necessitating:

$$\text{If } G, \text{ then: } \left[N \text{ because: } \left(G \wedge (G \underset{C}{\Rightarrow} N)\right)\right].$$

[6] Of course, if some φ obtains, we may be able to explain its obtaining because *all* of its constituents also obtain in the right configuration (assuming that φ is not primitive in a strong sense, see §4.4, below). But this is *not* what I am calling 'constitutive explanation'.

Therefore, constitutivism is compatible with the claim that the necessary connections between grounds and groundees *are due to grounding*—i.e., $\text{due}_{\text{constitutive}}$ to grounding. And, in contrast to groundism about necessity, (i) the explanation does not stop *there*: N is $\text{due}_{\text{constitutive}}$ to G because part of what individuates grounding is its necessitating constitution, and this further helps to explain grounding being such a tight determinative relation.

I ask the reader to consider whether this explanation is at least as attractive as the explanation purportedly provided by the groundist's **EN**.

4.4. Objection: Necessitation and primitive ground

Many grounding theorists hold that grounding is *primitive*. But if grounding is primitive, it cannot have a constitution. Then, constitutivism must be false.

A possible response is that grounding is not, after all, primitive: it is, at least, constitutively necessitating. As I remarked, important ground theorists do think of necessity as constitutive of grounding. Perhaps they are wrong—but perhaps, also, the view that grounding is primitive is wrong.

However, a more conciliatory view is possible. We can distinguish two senses of primitivity:

- Primitive_1 = being irreducible, not definable away in more fundamental terms, not having a nature exhaustively given by other phenomena;
- Primitive_2 = admitting no informative real definition or essential characterisation whatsoever, not having a nature given by other phenomena in any respect.

Constitutivism is inconsistent with grounding being primitive_2, but not obviously with grounding being primitive_1. The crucial point is about the *exhaustivity* of the constitutive requirements.

For example, the nature of pair-formation in the standard presentation of ZFC set theory is exhaustively defined in terms of sets through the Kuratowski definition: for any A, B, what it is for there to be its ordered pair $\langle A, B\rangle$ is exhaustively defined by there being a set of the form: $\{\{A\}, \{A, B\}\}$. Contrast with identity: it is not incoherent to take it as being essentially reflexive, symmetric, etc., but also primitive_1, as it does not have a nature *exhaustively* given by other phenomena. Similarly, grounding can be irreducible while being such that its constitution entails various facts—at least, an N-fact for each G-fact. So, grounding can have a necessitating constitution without thereby having an exhaustive definition.

5. Grounding to Fundamentality

5.1. The standard connection

An important assumption in the metaphysical research on fundamentality is the thought that A is more fundamental than B if A *determines*, *builds*, *constitutes* or *explains* B (or is the *kind* of thing that does so with respect to B). The entities related by fundamentality have been variously thought to be facts, propositions, objects, properties, or objective differences corresponding to ideologies (see Bennett, 2017, ch. 2, for an overview).

Whatever kind of entities they may be, A may be more fundamental than B even if it is not a foundation *for* B: A need only be as fundamental as something, call it 'C', that *does* constitute (etc.) B (Correia, 2021b). In this case, A would be of the same ontic 'level' or 'layer' than C. This notion that reality is separated into fundamentality levels leads to the *relative* notion of fundamentality, where some entities are more fundamental than others. We also have an *absolute* notion of fundamentality: to a first approach, the *absolutely* fundamental would be what is not determined (etc.) by anything else, or by being part of what determines (etc.) everything else (Tahko 2018; Bennett 2017, ch. 5, emphasises the first conjunct; see also Leuenberger 2020). Sometimes the fundamental is also characterised as *freely recombinable*, so that the space of all ontic possibilities is determined by all the recombinations of the fundamental ontology, but see Wang, 2016.

In some approaches, fundamentality is supposed to come in different varieties, which correlate with various assumptions as to what kind of *dependence* is at work. It has been widely assumed that dependence is tied to *explanation*: the dependent objects should be explainable by what they depend on—which, in turn, should be more fundamental. In this way, fundamentality, dependence, and explanation form 'a certain kind of trinity' (adapting the triad originally proposed by Schnieder, 2006.)

Most of these notions are related to some *building relation*, as Bennett (2017) calls them. An important principle of her view is that if x at least partially builds y, then x is more fundamental than y (Bennett, 2017, chs. 3, 6). Given that she takes grounding to be a building relation, her principle entails **GTF**. In fact, she takes each building relation to define an order of relative fundamentality (see §6.3, below). For our purposes, note that a very popular assumption is that grounding—with its attendant kind of explanation, and its intrinsic link to fundamentality—is one of the more (if not the most) important building or generative relation operative in the problems of metaphysics.

Indeed, one of the theoretical roles for which grounding was introduced was to *account for* the existence of the derivative as what is *ultimately grounded in the fundamental* (Fine, 2001; Schaffer, 2009). As Miller (2022, p. 1) says,

> One strong motivation for doing our metaphysical theorizing in terms of grounding is its promise to satisfactorily explicate the idea of reality having a layered structure. On this view, the layered structure of reality is given by a specification of which facts ground others.

This is, then, the *standard connection* between grounding and fundamentality: the more fundamental facts explain less fundamental ones by grounding them, while less fundamental facts depend on more fundamental ones by being grounded in them.

The standard connection is helpful for the grounding theorists when rebutting a certain kind of objection. Koslicki (2015) has argued that the purely formal properties of grounding are insufficient to distinguish it from other relations. Grounding theorists cite their stock examples, but Koslicki argues that they are much less unified than commonly thought (see also Turner, 2016 and Wilson, 2014; relatedly, McSweeney, 2020 and Merlo, 2022 argue that the logic of grounding traces semantic intuitions whose link to ontological dependence is tenuous.) As part of a response to these criticisms, the grounding theorists should argue that what distinguishes grounding from other formally similar relations, and what unifies the apparently dissimilar cases, is that grounding *generates less fundamental facts from more fundamental ones*. The standard connection provides material content to the bare grounding formalism.

GTF enunciates a link in the standard connection: every case of grounding is an explanatory connection between fundamentality levels—specifically, such that the grounds are more fundamental than their grounded facts. And because grounding was introduced to *provide*, or *back*, explanatory links from the fundamental to the derivative, there *must* be a non-empty class of cases where grounding entails fundamentality.

Below I introduce a theory on which the grounding order and the generative order come apart. This will require rejecting unrestricted **GTF**, but will yield a more principled grounding-fundamentality connection that survives the counterexamples that refute **GTF**.

5.2. Objection: Grounding-distance and disjunctions

Building on a case by Bennett (2017, p. 151), Correia (2021a, p. 5973) presents a case against **GTF** (see also Vogt, 2026, p. 2824, ftnt. 34; Bennett, Correia and Vogt note that **GTF** is fairly plausible and popular, notwithstanding). Using Vogt's formulation:

> the relative grounding-fundamentality of a 'fact' corresponds to [...] the number of steps of immediate ground that is needed in order to arrive at the fact from the fundamental level.

Then,

> Take the disjunction of some fundamental ϕ and an unrelated, highly non-fundamental ψ. On the orthodox view, $\phi \vee \psi$ is grounded in ψ. Yet, one can arrive at $\phi \vee \psi$ in just one step of immediate ground from the fundamental fact ϕ, while, by assumption, it takes many steps to arrive at ψ. So, on the account, $\phi \vee \psi$ is more fundamental than ψ despite being grounded in it [...]

The case establishes an incompatibility between **GTF**, the existence of disjunctive facts, the orthodox logic of grounding, and the 'shortest grounding genealogy'-conception of relative fundamentality.

I am not convinced of the existence of disjunctive facts. I *am* convinced of the existence of disjunctive propositions; but why think that, just after the Big Bang occurred, there was already a *fact* such as *Either quantum field q is in configuration C at spacetime region r, or there is a shrimp taco on my plate*? Further, Correia defines T-rank as the smallest T-height of a complete grounding tree, but gives relatively little metaphysical interpretation of why the shortest genealogy—rather than longest genealogy or some other function of the set of genealogies—should determine fundamentality. For the sake of discussion, however, I will not pursue these avenues. I will rather suggest a substitute of **GTF** that is motivated by a *generative conception* of fundamentality.

Let 'r' be the grounding-rank function, and 'D' the disjunction $(\phi \vee \psi)$. For illustration, suppose $r(\phi) = 1$ and $r(\psi) = 100$. Then, ϕ grounds D at $r = 2$, and ψ grounds D at $r = 101$. So, according to the argument, $r(D) = 2$, but ψ at $r = 100$ is a ground of it, so the ground is less fundamental than its groundee.

Well, *a* ground. But then, why is $r(D) = 2$ rather than 101? There would seem to be some arbitrariness in choosing the *minimal* rank at which F is grounded as F's grounding-rank. Fortunately, the suggested substitute of **GTF** will answer this. To explain how, let me explain my understanding of fundamentality.

5.3. The generative-predicative conception of fundamentality

On the *generative* conception of fundamentality, the fundamentality order is given by the stages in which the kinds of worldy phenomena emerge into being. These *stages* need not be temporally successive (although some of them may be): on the usual conception, the stages of set construction are not temporal; or if ψ grounds D, the stage at which ψ occurs differs from the stage at which D occurs, even if these facts obtain simultaneously.

On this conception, the *ordo essendi*, the fundamentality order, is the order of *coming into being.* At least some entities require that other entities combine or be operated upon in a certain way to come into being. Therefore, the *ordo essendi* is the order of generation:

Generative conception of fundamentality: The *ordo essendi* is the *ordo generationis*: the order between

the stages at which the less fundamental is generated from, or built out of, the more fundamental.

The generative conception of fundamentality is consistent with the claim that each of the generators carry their own explanatory power; relative fundamentality then records priority in the generative-explanatory order. But note: the view is not committed to metaphysical foundationalism: generation may or may not require absolute fundamentality.

Then, if x is more fundamental than y, then x is prior in the *ordo essendi* to y and so, it cannot be that y is used to somehow generate x, to somehow build x into being. This is a metaphysical analogue of *predicative construction*: an item at a stage cannot be generated using something that is not already available before that stage. Thus,

Metaphysical-generation predicativism: Reality at a stage of the *ordo generationis* can be generated only from what is already available before that stage.

What is generated acquires its place in the hierarchy by being generated; later material cannot retroactively become part of what generated it.[7] Then, *generation* fixes priority, but *later alternative sufficiency* does not.

Thus, we can distinguish two roles that grounding may play in the generative order. First, as *generative sufficiency*: if $\Delta > \varphi$, then Δ constitutes a sufficient metaphysical basis for φ. Second, as *operative generation*, where an *immediate* grounding tie may be the one through which φ first enters the *ordo generationis*. The crucial idea for replacing **GTF** is that these roles can come apart. A fact may have many complete grounds, including grounds available only after the fact has already been generated; such grounds remain metaphysically sufficient, but they do not determine the fact's place in the generative order.

Consider again the disjunction case. On the present conception, since ϕ is already available at $r = 1$, the immediate-grounding tie $\phi >_i D$ generates D at the next generative rank. When ψ becomes available much later, $\psi >_i D$ is still a genuine grounding connection, but it does *not* generate D into being *again*: by then D is already available. The latter tie therefore establishes generative sufficiency without fixing D's first position in the *ordo essendi*.

This explains why the minimum grounding genealogy is privileged: because generative rank represents *first availability*. Once a fact has entered the generative order, later sufficient derivations cannot retroactively determine where it entered. In this respect, **GTF** conflates two roles of grounding: providing a sufficient metaphysical basis and actually operating in the first generation of the grounded fact.

We can therefore define the relevant privileged grounding relation in terms of *first grounds*. Begin with fundamentality as a strict generative order: '$x \gg y$' means that x is available prior to y in the *ordo generationis*. This order need not have minimal elements. Then, a full ground Δ of φ is among the *first*

[7] Metaphysical predicativism is not metaphysical *finitism*. If the constitution of x requires an infinite family $\{y_i\}_{i \in I}$, there need not be a largest prior stage: x may arise at a limit or successor stage after the entire relevant basis is available.

grounds of φ ('$\Delta \underset{\text{first}}{>} \varphi$') iff $\Delta >_i \varphi$ and Δ is minimal, with respect to generative priority, among the full immediate grounds of φ: there is no $\Gamma >_i \varphi$ generatively prior to Δ (where a plurality becomes available once all of its members are available). Note that φ may have different first grounds.

This yields the new principle:

Generative GTF (gGTF): If $\Delta \underset{\text{first}}{>} \varphi$, then, $\forall \delta \in \Delta : \delta \gg \varphi$.

Unlike unrestricted **GTF**, **gGTF** does not require every sufficient ground to be more fundamental than its groundee. It requires this only of the first grounds through which the groundee enters the *ordo generationis*. Later grounds may remain perfectly genuine grounds while being redundant with respect to generation.[8]

We can introduce numerical, Correia-style r-ranks when the generative ordering admits such a ranking (for example, assuming that there are fundamental facts); an example construction is presented in the Appendix. There may be several such first grounds: distinct equifundamental grounding ties may generate the same fact at its first stage of availability. And under natural assumptions, **gGTF** becomes a theorem of the generative conception of fundamentality.

Going back to the Correia case, then, what the example establishes is the failure of the inference from arbitrary full grounding to generative priority. It does not establish a counterexample to **gGTF**. If ϕ first generates D while ψ becomes available only later, then $\phi \underset{\text{first}}{>} D$, whereas $\psi > D$ without $\psi \underset{\text{first}}{>} D$. The latter relation expresses metaphysical sufficiency, but not operative generation. And **gGTF** flows from an independently motivated conception of fundamentality as providing the *ordo generationis*, which is naturally paired with a *predicativity constraint on generation*. Thus, rather than being an *ad hoc* exception for disjunction, **gGTF** flows from a general metaphysical stance: the *generative-predicative* conception of fundamentality.

5.4. Objection: Multiple senses of grounding and grounding without fundamentality

The disjunction case targeted the grounding-fundamentality link by claiming that under the relevant sense of 'grounding', not all grounds are more fundamental than their groundees. Another objection may be that there are senses of 'grounding' where grounding need not connect what is more to what is less fundamental.

A number of philosophers have distinguished different senses of grounding and dependence (*e.g.*, Koslicki, 2015; Kovacs, 2018; Richardson, 2020; Turner, 2016; Wilson, 2014). Also, some have defended

[8] The distinction also permits a second kind of redundancy. If $r_g(\phi) = r_g(\psi) = 1$ and both independently immediately ground $\phi \vee \psi$, both grounding ties may be first grounds. Each is redundant in the sense that the other is sufficient, but neither is *generatively posterior*. We should therefore distinguish *late redundancy*, where a ground becomes available only after the groundee has already been generated, from *co-first redundancy*, where several equifundamental grounds are independently sufficient. Only the former prevents a grounding tie from fixing the groundee's generative rank.

the possibility or even actuality of dependency and/or grounding cycles (e.g., Bliss, 2014; Thompson, 2018; Morganti & Calosi, 2021; for discussion, see Busse, 2025). Thus, if φ grounds ψ, which grounds φ, then surely φ is not more fundamental than ψ nor ψ more fundamental than φ.

Now, *first*, I think that if grounding and fundamentality are *never* connected, this helps only to make the notion of grounding more obscure and less transparently applicable, and it would be doubtful that we are still using the term for what metaphysicians intend to mean. So, I assume that grounding and fundamentality are *at least in many cases* connected; and I have argued that the connection is given by **gGTF**. *Second*, note that I need not *prohibit* the existence of grounding cycles. For my argument to work, we need only to accept that **gGTF** holds for a set of cases that covers *G*- and *N*-facts.

And *thirdly*, whatever exotic senses of grounding may violate **gGTF**, the question is whether De Rizzo's explanatory 'because' is one such sense. Here, note that the central claim of many purported explanations or reductions of modality concerns the *source* of necessity. De Rizzo (2020, p. 639) explicitly asks 'where the necessity of necessary claims 'comes from', that is, why are necessary truths necessary?' And his account states that 'the source of necessity lies in explanations of particular forms [...]' He also claims that '**EN** suggests an 'explanation-first' account of the source of necessity', where necessity '*is owed*' to some grounding connections (p. 641). Taken at face value, this language strongly suggests a *generative conception* of the grounding order, where the grounds *source* the grounded.

But the *generative conception* of the grounding order does not pair well with failures of **gGTF**. In those cases, *N* and *G* would be either equally fundamental or incomparable as to their fundamentality. Some may find it somewhat interesting that grounding grounds necessity, even if neither is more fundamental than the other. However, this would not be useful to satisfy the desiderata that are sought when one proposes a theory about what is more generatively fundamental than what—for example, that the grounded is a derivative aspect of reality, which reduces our fundamental commitments; or that the 'coming from' figure means literally that the *sourced* aspect emerges into being.

Therefore, I do not think that groundists have a good prospect of rejecting **gGTF** while preserving the significance of their account.

5.5. The argument, again

gGTF will only do the work that **GTF** did in the incompatibility argument of §2 if the relevant grounds there are *first* grounds. And they are.

De Rizzo has two commitments. First, at the level of his *advertised interpretation* of the theory, the claim that grounding facts are the *source* of the necessity of necessary facts. Second, the implementation of this source claim in the final theory. There, **EN** generalises from $G > N$. Under the generative-predicative

conception, we can now understand the source claim as the claim that G-facts are what bring N-facts into the *ordo essendi*, rather than merely a later sufficient ground of an already generated fact. Let '**gEN**' be this interpretation of **EN**. Thus, the corrected argument does not purport to *derive* **gEN** from bare **EN**; it targets **EN** under the source interpretation with which De Rizzo motivates it:

1.	$G = (\Delta > p)$.	
2.	$N = \Box(\bigwedge \Delta \rightarrow p)$.	
3.	$G \underset{C}{\Rightarrow} N$.	**NC**
4.	$\therefore \neg(G \gg N)$.	**CTAF**, 3
5.	$G \underset{\text{first}}{>} N$.	**gEN**, 1, 2
6.	$\therefore G \gg N$.	**gGTF**, 5
7.	$\bot$.	

The groundists can preserve bare **EN** by claiming that, while G grounds N, N was already generated from some earlier ground. But then, G would only be a later, redundant ground. And if so, G would not be the generative source of N.

6. Constitution to Anti-Fundamentality

I have relied on the following principle for the central argument:

CTAF: If φ requires ψ because of its metaphysical constitution, then φ is not more metaphysically fundamental than ψ:

$$\text{If } (\varphi \underset{C}{\Rightarrow} \psi), \text{ then } \neg(\varphi \gg \psi).$$

Note that **CTAF** is importantly weaker than the claim that constitutive entailment implies strictly more fundamentality:

$$\text{If } (\varphi \underset{C}{\Rightarrow} \psi), \text{ then } (\psi \gg \varphi).$$

Given that $\varphi \underset{C}{\Rightarrow} \psi$, **CTAF** permits three possibilities: ψ is prior to φ, they are at the same level, or they are incomparable. **CTAF** rules out only the structurally peculiar possibility that the constituted fact is prior to one of the facts required by its constitution. And I think that this weakness is a major virtue: as we will see below (§6.2), it permits recursive definition, definitional cycles, mutually constituted items and structuralist cases. But first, let me explain why I think **CTAF** is true.

6.1. Generation and predicativity

CTAF essentially says that the arrows of fundamentality and constitutive requirement never point in opposite directions. This idea is motivated by three further ideas.

First, the *generative-predicative conception* of fundamentality (§5.3 above). Second (and influenced by Fine, 1991), I understand constitutive requirements as displaying 'metaphysical presuppositions': the constitution of x requires what it presupposes. Third, the thesis that, to be constituted, an entity presupposes both what generates it (if anything), and the constitutive structures (if any) that may be holistic or mutually specified. The former, through the predicativity of generation (§5.2, above), leads to more fundamental things; the latter allows equifundamental entities. What never happens is that a thing's constitution metaphysically presupposes something that is not already present at the stage at which the thing is constituted.

Thus, given the understanding of constitutive requirements as detailing metaphysical presuppositions, if x's constitution requires y, then y has to be given at a generative stage not later than the stage at which x emerges. But if y is ontologically *posterior* to x, then x appears in the *ordo essendi before* y. Then, a metaphysics denying **CTAF** owes us an account of how the constitution of an item can require something *ontically posterior* to it.

To sum up, the argument is this. Under the generative conception of fundamentality, the order of priority is the *ordo generationis*. Constitutive requirements say what a fact's constitution presupposes as given. But if $\varphi \underset{C}{\Rightarrow} \psi$ and φ is strictly more fundamental than ψ, then φ is generated at a stage of the *ordo essendi* prior to that of ψ. Then, if **CTAF** fails, a fact's constitution requires something that is not already available. But nothing can obtain without what it presupposes. Therefore, **CTAF** cannot fail.

And **CTAF** is not peculiar to the present case. Analogous non-reversal constraints appear wherever constitution is paired with a priority ordering: semantic composition, recursive specification, and familiar cases of metaphysical constitution all exhibit the same pattern. This does not independently prove **CTAF**, but it shows that the principle instantiates a broader predicative structure.[9]

For example, suppose a complex property P is metaphysically constituted, at least partly, by a property S. Then it is very natural to say that it follows that P is not more fundamental than S. If a conjunctive property is a genuinely complex property, it would be extraordinary if the conjunctive property were metaphysically prior to one of the properties constituting it. Or consider semantic constitution and priority. Suppose that A's meaning is compositionally determined partly through B's meaning; thus: $[\![A]\!] = f([\![B]\!], [\![C]\!], \ldots)$. It would therefore be bizarre to say that A's meaning is *semantically more primitive* than B's. But it does *not* follow that B must be strictly more primitive than A: there may be mutual definability, simultaneous specification, or holistic determination. What follows is just that, if the semantic constitution of A's meaning requires B's meaning, then, it is not the case that A's meaning is more

[9] According to priority monism (Schaffer, 2010), the cosmos is composed of proper parts while nevertheless being metaphysically prior to those parts. But this only means that if priority monism were true, mereological composition would, therefore, not follow metaphysical constitution requirements so as to obey **CTAF**. In a priority-monist universe, the cosmos' constitution does not require its parts: x being a part of y does not imply $y \underset{C}{\Rightarrow} x$.

semantically primitive than B's meaning.

I will now consider two potential objections to **CTAF**.

6.2. Objection: Circles, recursivity, and impredicativity

It may be argued that some things cyclically, symmetrically, or circularly require one another. I need not quarrel: **CTAF** only says that what is constitutively required by x cannot be metaphysically *posterior* to x itself. It is compatible with symmetries, cycles and circles. Also, there are cases where an object seems to be individuated in terms of—and thus, constitutively require—objects not prior to it. In more detail, let us consider structuralism, as well as impredicative and recursive definitions.

Structuralist philosophers of mathematics and physics have proposed identity criteria whereby some objects are exhaustively individuated by their mutual relations (MacBride, 2006; Ladyman, 1998; Muller, 2015). ***CTAF** is compatible with structuralism:* **CTAF** only entails that neither of the *relationals* is more fundamental than the other, and this is compatible with structuralism.

Recursive definitions sometimes are given in terms including the *definiendum* itself. Can an object be individuated by a recursive definition? (*cf.* Rosen, 2015, p. 196). ***CTAF** is compatible with recursive definitions:* It only entails that the *definiens* is not more fundamental than itself—which is true. Even more, recursive definitions can *look* circular without *being* viciously circular. Consider this recursive definition of the action of a function f:

$$f(0) = a, \qquad f(n+1) = F(f(n)),$$

for some condition F. Then, f occurs in its own definition. However, the real chain of dependency is well-founded:

$$f(0) \prec f(1) \prec f(2) \prec \cdots$$

For no n is $f(n)$ dependent on itself or on later values. Surface self-reference is therefore compatible with generative priority, given the well-foundedness of recursive definitions.

Finally, impredicative definitions mention something that is generatively posterior to the thing being defined (for example, the definition of a set S in terms of another set that contains S). ***CTAF** is compatible with impredicative definitions—under an interpretation.* First, **CTAF** is *not* compatible with the following thesis: *an impredicative definition in terms of a set that contains the definiendum gives the metaphysical requirements of the definiendum.* This, at least if we assume (and I do) that the elements are generatively prior to the sets they constitute. In the history of mathematics, ideas related to this have been used to reject impredicative definitions, but the fact that they are disqualified by **CTAF** as enunciating constitutive requirements does not impinge their mathematical status—under the assumption that there is some other, appropriately

predicative account of the *definiens*' mathematical generation.[10]

6.3. Objection: Multiple fundamentalities

Finally, another potential objection is that I have equivocated between different senses of 'fundamentality'. Then, there should at least be two different notions of fundamentality: one entailed by grounding, but not by constitutive requirement; another entailed by constitutive requirement, but not by grounding. For example, if there is a family of building relations and at least some of them entail their own fundamentality order (Bennett, 2017), *then*, **gGTF** may hold (at most), for the fundamentality limned by grounding, and the challenge would be to show that **CTAF** limns the same order.

This uncovers an assumption of my main argument (§2): if there is no univocal fundamentality order, there is no contradiction. Thus, I am committed to:

Unity Constitutive requirements and grounding are constraints on a single metaphysical fundamentality ordering, $\gg$.

Distinguish three views:

Building and Fundamentality Singularism There is only one way of generating the less fundamental from the more fundamental, and thus a single relative fundamentality ordering (*cf.* Bertrand, 2017).

Indexed Pluralism Different building relations generate different legitimate relative fundamentality orders (*cf.* Bennett, 2017, ch. 6).

Building Pluralism And Fundamentality Singularism There are multiple dependence and building relations, but they are all constraints on a single underlying fundamentality order (*cf.* Wilson, 2025).

The third approach is also my view, and it stems from the generative-predicative conception of fundamentality: *fundamentality* is the generative order of reality, where entities at a stage never metaphysically

[10] An illustrative example is the definition of the union operation in Zermelo's set theory:

Union $U(S) = \{x : (\exists Y)(Y \in S \wedge x \in Y)\}$

As Feferman (2005, p. 597) explained, Poincaré criticized this 'on the grounds that what members $[U(S)]$ has, depends on which sets belong to S and what their members are'. The problem is that, if $U(S) \in S$, then $U(S)$'s definition appeals to a set (i.e., S) that contains it, and this is assumed to entail that $U(S)$ is strictly prior to S. However, even in *that* case, **CTAF** is consistent with **Union**—it only disqualifies **Union** as giving the constitutive requirements of $U(S)$. After all, if $U(S) \in S$, then $U(S)$ appears in the cumulative hierarchy *before* S. Then, $U(S)$ is constituted by sets appearing previously in the hierarchy; so, **Union** need not give (in extension) the true constitutive requirements of $U(S)$.

presuppose what is available only later. Then, two generative relations can differ, but if both are *truly* generative, their directional implications must be jointly embeddable in the same generative structure.[11]

Now, what seems clear is that different kinds of entities are differently generated—so, perhaps *building* contains several species of relation (as Bennett, 2017, ch. 2, thinks). Also, different relations can have different formal behaviour: being unary or plural; etc. But there is at least one unifying feature: *generative relations may use only already available materials.* To the extent that builders are generative, all builders are metaphysically predicative.

However, if the indexed fundamentality relations of the pluralist are mutually independent, nothing prevents cases where x fully B_1-builds y, but y fully B_2-builds x. But how can something fully build things from which it itself is fully built? We seem to require 'harmony' principles—and indexed pluralism does not seem to especially motivate any. In contrast, a harmony principle is naturally motivated by the generative-predicative conception: no cross-index fully-building cases are possible because all building relations partake of the single stream of existence, of the single *ordo essendi.*[12]

Also, an output of B_1 can itself be an input to B_2: e.g., a biological fact grounds some psychological fact, and set formation builds the singleton of the latter. But if we can compose building relations, then there must be a meaningful answer to: *Is x available yet for B_2 to use?* And this requires a *relation-neutral* notion of availability—which takes us most of the way to the global *ordo generationis.*

Thus, I reject indexed pluralism not by merely insisting that there be a unique fundamentality order, but by suggesting an account of such uniqueness: *fundamentality is the global availability order presupposed by all generative relations*, because it is the *ordo generationis* that ensures *metaphysical predicativity.*

Even leaving the above to the side, indexed pluralism is not a true ally of groundism. As I remarked above, many purported explanations or reductions of modality, groundism included, concern the *source* of necessity—they are typically *generative* (*cf.* Wallner, 2022). But if G grounds N, while G can only be metaphysically constituted once N is already available, in what sense is G genuinely the *source* of N?

Then, the pluralist needs to offer a competing conception of what a metaphysical 'source' is, if source relations can run in both directions between the same two facts *according to distinct axes.* And if both arrows genuinely describe generation—if source and sourced mutually presuppose one another via distinct fundamentality orders—there remains a question indexing does not answer: *What is the overall generative*

[11] This third approach need *not* equal Bennett's *sophisticated primitivism* (2017, §6.2.2), on which the relative fundamentality facts are primitive, and *constrain*, i.e. '*enable* and *prohibit* various patterns of building' (p. 182). What is true is that relative fundamentality and building relations constrain each other. But this mutual constraining may indicate that relative fundamentality and building generate each other (Bertrand, 2017, p. 1281), or perhaps that they share a common source.

[12] Even if—following Bennett (2017, ch. 5)—the indexed pluralist defines *absolute fundamentality* as *not built by any builder*, this only provides a single common base, but does not imply much about cross-index fully-building for derivative entities.

architecture containing both arrows? As we see, the indexed-pluralist groundist has moved very far from the advertised picture on which grounding provides *the source* of necessity.

7. Conclusion

According to the groundist conception of necessity, the necessity of necessary facts is explained by their being grounded in grounding facts. De Rizzo (2020) provides a systematic development of groundism.

I have proposed an alternative picture, which is strictly inconsistent with how De Rizzo advertises the central principle of groundism, **EN**. The first major component of this picture is the thesis that grounding is constitutively necessitating—which explains its *tightness*, separates it from other explanatory relations, and provides no less explanatory power than the groundist conception of necessity. The second major component is a metaphysics of fundamentality as the *ordo generationis*, where fundamentality is the global availability order presupposed by all generative relations. This order enforces the *predicativity* of those generative relations. From this conception, I have motivated a connection between grounding and fundamentality (and defused counterexamples to it), a connection of constitutional requirement to fundamentality, as well as the uniqueness of the fundamentality order.[13]

References

Bennett, Karen. 2017. *Making Things Up.* Oxford University Press.

Bennett, Karen. forthcoming. The Determination First Approach to Plexology. *Australasian Philosophical Review.*

Bertrand, Michael. 2017. Fundamental Ontological Structure: An Argument Against Pluralism. *Philosophical Studies*, **174**(5), 1277–1297.

Bliss, Ricki. 2014. Viciousness and Circles of Ground. *Metaphilosophy*, **45**(2), 245–256.

Bliss, Ricki, & Trogdon, Kelly. 2024. Metaphysical Grounding. *In:* Zalta, Edward, & Nodelman, Uri (eds), *The Stanford Encyclopedia of Philosophy*, summer 2024 edition edn. Metaphysics Research Lab, Stanford University.

Boolos, George. 1984. To Be is to Be a Value of a Variable (or to Be Some Values of Some Variables). *Journal of Philosophy*, **81**(8), 430–449.

Busse, Ralf. 2025. No Foundations for Metaphysical Coherentism. *Philosophical Studies*, **182**(9), 2503–2534.

Correia, Fabrice. 2021a. Fundamentality From Grounding Trees. *Synthese*, **199**, 5965–5994.

Correia, Fabrice. 2021b. The Logic of Relative Fundamentality. *Synthese*, **198**, 1279–1301.

Correia, Fabrice. 2024. *The Logic of Grounding.* New York, NY, USA: Cambridge University Press.

Daly, Chris. 2023. Explanation Good, Grounding Bad. *The Monist*, **106**(3), 270–286.

De Rizzo, Julio. 2020. Grounding Grounds Necessity. *Analysis*, **80**(4), 639–647.

Feferman, Solomon. 2005. Predicativity. *Pages 590–624 of:* Shapiro, Stewart (ed), *Oxford Handbook of Philosophy of Mathematics and Logic.* Oxford: Oxford University Press.

[13] [Acknowledgements]

Fine, Kit. 1991. The Study of Ontology. *Noûs*, **25**(3), 263–294.

Fine, Kit. 2001. The Question of Realism. *Philosophers' Imprint*, **1**, 1–30.

Fine, Kit. 2002. The Varieties of Necessity. *Pages 253–282 of: Conceivability and Possibility.* Oxford University Press.

Fine, Kit. 2012. Guide to Ground. *Pages 37–80 of:* Correia, Fabrice, & Schnieder, Benjamin (eds), *Metaphysical Grounding.* Cambridge University Press.

Glazier, Martin. 2017. Essentialist Explanation. *Philosophical Studies*, **174**(11), 2871–2889.

Haderlie, Derek Christian, & Miller, Taylor-Grey. forthcoming. Essence, Modality, and Hyperintensional Metaphors. *Synthese.*

Kment, Boris. 2014. *Modality and Explanatory Reasoning.* Oxford University Press.

Koslicki, Kathrin. 2015. The Coarse-Grainedness of Grounding. *Oxford Studies in Metaphysics*, **9**, 306–344.

Kovacs, David Mark. 2018. The Deflationary Theory of Ontological Dependence. *Philosophical Quarterly*, **68**(272), 481–502.

Ladyman, James. 1998. What is Structural Realism? *Studies in History and Philosophy of Science Part A*, **29**(3), 409–424.

Leuenberger, Stephan. 2014. Grounding and Necessity. *Inquiry*, **57**(2), 151–174.

Leuenberger, Stephan. 2020. The Fundamental: Ungrounded or All-Grounding? *Philosophical Studies*, **177**(9), 2647–2669.

Lo, Tien-Chun, Rodríguez-Pereyra, Gonzalo, & Skiles, Alexander. forthcoming. Against Zero-Grounding. *Philosophical Studies.*

MacBride, Fraser. 2006. What Constitutes the Numerical Diversity of Mathematical Objects? *Analysis*, **66**(1), 63–69.

McSweeney, Michaela. 2020. Debunking Logical Ground: Distinguishing Metaphysics From Semantics. *Journal of the American Philosophical Association*, **6**(2), 156–170.

Merlo, Giovanni. 2022. Disjunction and the Logic of Grounding. *Erkenntnis*, **87**(2), 567–587.

Miller, Taylor-Grey. 2022. Essentialist Non-Reductivism. *Philosophers' Imprint*, **22**(n/a).

Morganti, Matteo, & Calosi, Claudio. 2021. Interpreting Quantum Entanglement: Steps Towards Coherentist Quantum Mechanics. *British Journal for the Philosophy of Science*, **72**(3), 865–891.

Muller, F. A. 2015. The Rise of Relationals. *Mind*, **124**(493), 201–237.

Pearce, Kenneth. 2025. Metaphysical Rationalism Requires Grounding Indeterminism. *Journal of the American Philosophical Association*, **11**(2), 303–322.

Raven, Michael. 2015. Ground. *Philosophy Compass*, **10**(5), 322–333.

Richardson, Kevin. 2020. Grounding Pluralism: Why and How. *Erkenntnis*, **85**(6), 1399–1415.

Rosen, Gideon. 2006. The Limits of Contingency. *Pages 13–39 of:* MacBride, Fraser (ed), *Identity and Modality.* Oxford University Press.

Rosen, Gideon. 2010. Metaphysical Dependence: Grounding and Reduction. *Pages 109–36 of:* Hale, Bob, & Hoffmann, Aviv (eds), *Modality: Metaphysics, Logic, and Epistemology.* Oxford University Press.

Rosen, Gideon. 2015. Real Definition. *Analytic Philosophy*, **56**(3), 189–209.

Rosen, Gideon. 2017. Ground by Law. *Philosophical Issues*, **27**(1), 279–301.

Rydéhn, Henrik. 2022. Metaphysically Opaque Grounding. *Canadian Journal of Philosophy*, **52**(7), 729–745.

Schaffer, Jonathan. 2009. On What Grounds What. *Pages 347–383 of:* Manley, David, Chalmers, David, & Wasserman, Ryan (eds), *Metametaphysics: New Essays on the Foundations of Ontology.* Oxford University Press.

Schaffer, Jonathan. 2010. Monism: The Priority of the Whole. *Philosophical Review*, **119**(1), 31–76.

Schnieder, Benjamin. 2006. A Certain Kind of Trinity: Dependence, Substance, Explanation. *Philosophical Studies*, **129**(2), 393–419.

Skiles, Alexander. 2015. Against Grounding Necessitarianism. *Erkenntnis*, **80**(4), 717–751.

Tahko, Tuomas. 2018. Fundamentality. *In:* Zalta, Edward N. (ed), *The Stanford Encyclopedia of Philosophy*, fall 2018 edn. Metaphysics Research Lab, Stanford University.

Thompson, Naomi. 2018. Metaphysical Interdependence, Epistemic Coherentism, and Holistic Explanation. *Pages 107–125 of:* Bliss, Ricki, & Priest, Graham (eds), *Reality and its Structure: Essays in Fundamentality.* Oxford University Press.

Trogdon, Kelly. 2013. Grounding: Necessary or Contingent? *Pacific Philosophical Quarterly*, **94**(4), 465–485.

Turner, Jason. 2016. Curbing Enthusiasm About Grounding. *Philosophical Perspectives*, **30**(1), 366–396.

Vogt, Lisa. 2026. Upwards Essence. *Philosophical Studies*, **183**(9), 2805–2834.

Wallner, Michael. 2022. Essence, Necessity, and Non-Generative Metaphysical Explanation. *Argumenta*, **7**(2), 439–462.

Wang, Jennifer. 2016. Fundamentality And Modal Freedom. *Philosophical Perspectives*, **30**(1), 397–418.

Wilson, Alastair. 2018. Metaphysical Causation. *Noûs*, **52**(4), 723–751.

Wilson, Jessica. 2014. No Work for a Theory of Grounding. *Inquiry*, **57**, 535–79.

Wilson, Jessica. 2025. The Fundamentality First Approach to Metaphysical Structure. *Australasian Philosophical Review*, **1**(1), 1–34.

Wygoda Cohen, Shlomit. 2024. Indeterministic Grounding. *Synthese*, **204**(1), 1–21.

Appendix

I assume foundationalism for this illustration, as well as Correia's grounding-tree machinery (Correia, 2021a), with trees constructed from immediate (and hence non-transitive) grounding ties, and transcendent facts excluded. Let $>_i$ denote immediate full grounding. We can then define the *generative rank* of a fact recursively:

$$r_g(\varphi) = \begin{cases} 1, & \text{if } \varphi \text{ is fundamental,} \\ \min\limits_{\Delta >_i \varphi} \left(r_g(\Delta) + 1 \right), & \text{otherwise} \end{cases} \tag{1}$$

where, for pluralities Δ,

$$r_g(\Delta) = \sup_{\delta \in \Delta} r_g(\delta).$$

To justify the second clause in (1): once an immediate full ground is available, **Necessitation** entails that its groundee is also available, and predicative priority entails that $r_g(\Delta) < r_g(\varphi)$. Then, representing an immediate operative grounding step by a successor rather than by an arbitrary gap is the canonical way to preserve the absence of any intervening generative dependence.

First grounding can then be characterised by:

$$\left(\Delta \underset{\text{first}}{>} \varphi \right) \text{ iff } \left(\Delta >_i \varphi \text{ and } r_g(\varphi) = r_g(\Delta) + 1 \right).$$

And then, $r_g(\Delta) < r_g(\varphi)$. Therefore, since relative generative fundamentality is represented by rank: for every $\delta \in \Delta$, $\delta \gg \varphi$; this yields **gGTF**.